XLM-R-Based Glot500-Style Vocabulary Extension for Low-Resource Tail Languages

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Term Project Report Draft
July 2026

Abstract

XLM-R is a strong multilingual masked language model covering 100 languages, but it over-fragments tokens of unseen tail languages (tokens/word = 2.204) and fails to align their lexical and morphological features in the representation space. Glot500 addresses this by extending XLM-R's vocabulary with a mixed-corpus SentencePiece tokenizer and continuing pretraining via masked language modeling (MLM), and reports improved tail-language representations through related-language help. A key step Glot500 leaves open is how to initialize the new token embedding rows.

We investigate this choice with a controlled 5-way ablation on `xlm-roberta-base`. The tokenizer is fixed to Glot500-style SentencePiece append, adding 116,664 new pieces, and only the initialization of those new rows varies across five methods: `random`, `mean`, `fvt` (Fast Vocabulary Transfer), `weighted_fvt` (length-weighted FVT), and `family_mean` (family-aware mean). We train on 92 XLM-R-seen replay languages and 7 XLM-R-unseen target languages (Target7) for 50K steps under identical corpus, objective, and evaluation protocol. Evaluation follows the Glot500 downstream suite: pseudoperplexity (PPPL), Tatoeba and Bible sentence retrieval, NER, roundtrip alignment, and text classification.

We observe three findings. (1) The extended tokenizer reduces Target7 tokens/word from 2.204 to 1.592 (27.75% reduction), confirming that tail-language fragmentation is alleviated. (2) Initialization alone changes 50K convergence loss and downstream scores: `family_mean` achieves the lowest target PPPL (12.3) and highest Tatoeba tail accuracy (36.1), while `weighted_fvt` is strongest on training loss and roundtrip alignment. FVT-family methods lead at 10K steps but are overtaken by `weighted_fvt` and `family_mean` by 30–50K, showing that the benefit of structured initialization emerges at convergence, not at early checkpoints. (3) The sentence representation space exhibits language-identity and family/typology clustering, and languages with strong family coverage in the head set benefit most from `family_mean`. In short, embedding initialization is a consequential design choice for low-resource vocabulary adaptation, and corpus-provenance or family-prior initialization outperforms random initialization in the target regime.

1 Introduction

1.1 Vertical and Horizontal Scaling

Advances in large language models have proceeded along two axes. **Vertical scaling** deepens understanding of a fixed set of high-resource languages by increasing parameter count, training data, and compute—ChatGPT, Claude, and scaling-law research belong here. **Horizontal scaling** asks instead whether a model can support more languages, domains, and tasks. Supporting low-resource tail languages is the canonical horizontal-scaling problem and is the background of this report. XLM-R [Conneau et al., 2020] is a strong multilingual encoder, but for languages absent from its training data it suffers from two compounding deficits: (1) token over-fragmentation, where a single morpheme is split into many subword pieces, and (2) representation mismatch, where the embedding space carries no signal from the target language.

1.2 The Open Choice in Vocabulary Extension

Glot500 [Imani et al., 2023] combines vocabulary extension with continued MLM pretraining to address these deficits for 511 language-scripts. After extending the tokenizer, XLM-R's embedding matrix $\mathbf{E} \in \mathbb{R}^{250002\times768}$ grows to $\mathbb{R}^{366666\times768}$: the 116,664 new rows have never received a gradient. Glot500 initializes these rows randomly. Whether a different starting point —one that reuses information already encoded in the existing XLM-R embeddings—leads to better convergence and downstream performance under a limited training budget is the question this report addresses.

1.3 Research Questions

1. Does Glot500-style SentencePiece append reduce subword fragmentation for XLM-R-unseen tail languages?
2. With tokenizer and corpus held fixed, does changing only the initialization of new token rows alter PPPL and downstream task scores?
3. Does the sentence representation space exhibit language-identity or family/typology structure after continued pretraining?

1.4 Scope and Contributions

This experiment is not a full reproduction of Glot500 across 511 languages. All seven Target7 languages use Latin script, so we make no claim about script diversity. Instead, we select the smallest XLM-R-unseen tail-language corpus band that still supports downstream evaluation, and run a controlled 5-way initialization ablation with identical tokenizer, corpus, objective, schedule, and evaluation protocol.

Our contributions are:

1. We propose `family_mean`, a family-aware initialization that places new token rows at the centroid of source embeddings belonging to the same language family, and verify it against four baselines (`random`, `mean`, `fvt`, `weighted_fvt`) in a controlled 5-way ablation with the tokenizer fixed.
2. We connect the 2D similarity map's family clustering with downstream results, showing that family-aware initialization converges to the best target PPPL, Tatoeba retrieval, and NER scores.
3. We track the step-by-step formation of family geometry in the representation space and use it to interpret why family-prior initialization reaches a better optimum.

2 Related Work

2.1 Glot500 and Horizontal Language Scaling

Glot500 [Imani et al., 2023] is the horizontal-scaling approach most directly related to this work. It treats languages seen during XLM-R training as *head languages* and all others as *tail languages*, and scales the model to cover 511 language-scripts.

Corpus collection. Glot500-c aggregates text from religious texts (Bible translations), news articles, scientific papers, and web crawls. Data are organized by language-script unit and filtered with chunk-level filters (character repetition, word repetition, special characters, short texts, duplicates) and corpus-level filters (language–script mismatch, perplexity divergence against a 3-gram character LM). Only language-scripts with at least 30,000 training sentences are retained.

Tokenization. Glot500 trains a SentencePiece unigram tokenizer [?Kudo and Richardson, 2018] on Glot500-c with temperature sampling ($\alpha = 0.3$) and merges the resulting vocabulary into XLM-R's existing SentencePiece model by appending pieces not already present. Byte fallback prevents unknown tokens.

Continued pretraining. Glot500-m is produced by continued MLM pretraining of XLM-R-base on Glot500-c. Language sampling uses $\alpha = 0.3$ to prevent high-resource head languages from dominating.

Key findings. Glot500 reports that performance depends jointly on corpus size, script coverage, related-language support, and model capacity. Notably, having related languages in the training set improves tail-language representations (*related-language help*). Our `family_mean` initialization directly encodes this prior into the new embedding rows.

2.2 Two Vocabulary Extension Approaches

Two vocabulary extension designs appear in the literature. The *mixed-corpus* approach (Glot500-style) trains a new tokenizer on a corpus of both seen and unseen languages and appends all new pieces to the source vocabulary, resulting in a large number of new tokens (116,664 in our experiment). The *target-only* approach [Yamaguchi et al., 2025] trains an auxiliary tokenizer on a small amount of target-language text (≈0.01 GB) and appends only the target-specific pieces, keeping the vocabulary increment small. This report fixes the main axis to Glot500-style mixed-corpus injection and compares only initialization; the Yamaguchi-style approach is a separate axis not varied here.

2.3 Embedding Initialization for New Tokens

Placing new vocabulary rows in the correct region of an existing embedding space has been studied under the name *vocabulary transfer*.

FVT [?] decomposes each new token's surface form with the source tokenizer and initializes the row as the mean of the constituent source embeddings. Our `fvt` implements this directly.

WECHSEL [Minixhofer et al., 2022] computes similarity between new tokens and source tokens using static multilingual word embeddings and initializes new rows as a weighted combination of similar source rows.

FOCUS [Dobler and de Melo, 2023] uses anchor tokens that overlap between source and target vocabularies to express new tokens as combinations of anchors.

Both WECHSEL and FOCUS demonstrate that reusing semantic or surface similarity beats random initialization—the same motivation behind `family_mean`, which reuses typological similarity. We do not implement WECHSEL or FOCUS because they require external static embeddings or separate anchor computation; we cite them as background.

Mean initialization [Hewitt, 2021] places new rows near the centroid of existing embeddings to reduce early instability, which is our `mean` baseline.

2.4 Evaluation Protocol

We follow the Glot500 downstream suite: PPPL [?], Tatoeba and Bible sentence retrieval [?], NER on WikiANN [?], roundtrip word alignment [?], and text classification on Taxi1500 [?]. Since labeled training data does not exist for tail languages, all supervised tasks fine-tune on English and evaluate zero-shot on target languages, following the XTREME protocol [?]. This setup ensures that cross-lingual alignment—the quantity that initialization directly affects—is the only pathway to tail-language performance.

3 Data

3.1 Language Scope and Data Sources

Our corpus universe consists of 99 language-scripts drawn from Glot500-c: 92 *head* language-scripts that XLM-R was trained on, and 7 *tail* language-scripts that XLM-R did not see (Target7). Both groups use Glot500-c as their text source, accessed as HuggingFace Arrow datasets (`{lang}_{script}`, `train` split, `text` column).

Head languages serve as replay data: including them in continued pretraining prevents catastrophic forgetting of XLM-R's existing representations. Target7 languages are the evaluation targets. They satisfy four selection criteria: (i) absent from XLM-R training, (ii) present in Glot500-c, (iii) at least 30,000 training sentences (`new_length` $\geq$ 30,000), and (iv) at least one downstream evaluation dataset available.

Table 1: Target7 languages with family, region, corpus size, and covered downstream tasks.

language_script	Name	Family	Region	new_length	Tasks
`dtp_Latn`	Kadazan Dusun	Austronesian	SE Asia	48,468	Tatoeba, Bible, Roundtrip, Taxi, Embe
`xav_Latn`	Xavante	Macro-Je	S America	31,765	Bible, Roundtrip, Taxi
`bam_Latn`	Bambara	Mande	W Africa	32,150	Bible, Roundtrip, Taxi
`csb_Latn`	Kashubian	Slavic	Europe	33,743	Tatoeba, NER, Embed.
`ile_Latn`	Interlingue	Constructed	Europe	40,984	Tatoeba, Embed.
`lij_Latn`	Ligurian	Romance	Europe	42,447	NER
`fur_Latn`	Friulian	Romance	Europe	30,052	NER

3.2 Task-Centered Language Selection

The seven languages were selected in reverse from the evaluation tasks, not arbitrarily. Most genuinely low-resource languages lack labeled downstream data, making it impossible to complete the tokenizer $\rightarrow$ training $\rightarrow$ evaluation pipeline end to end. We therefore required at least three evaluable target languages per downstream task, as shown in Table 2. Geographic and typological spread (SE Asia, South America, West Africa, Europe) was also considered, but traceability—being able to connect a specific score to a specific language—was the primary criterion.

Table 2: Task coverage: three Target7 languages per downstream task.

Task	Target languages	Scale
Tatoeba retrieval	`dtp`, `ile`, `csb`	2,253 sentence pairs
Bible retrieval	`dtp`, `xav`, `bam`	23,238 sentence pairs
Roundtrip alignment	`dtp`, `xav`, `bam`	22,669 sentence groups
NER	`csb`, `lij`, `fur`	300 sentences (100/lang.)
Embedding similarity	`csb`, `dtp`, `ile`	2,100 sentence pairs

3.3 Training Corpus and Temperature Sampling

The MLM training corpus mixes head replay and Target7 text. To prevent high-resource head languages from dominating the vocabulary and gradient signal, we apply temperature sampling with $\alpha = 0.3$:

$$P_i \propto n_i^{\alpha}, \quad \alpha = 0.3,$$

where n_i is the corpus size of language-script i. At $\alpha = 1$, a head language with one million sentences would be sampled 1,000× more often than a tail language with 1,000 sentences; at $\alpha = 0.3$ the ratio shrinks to approximately 8:1. This downweighting applies both to tokenizer training and to the actual MLM corpus.

Table 3: Training and evaluation corpus sizes.

Item	Count
Source raw sentences	1,025,895,043
MLM train — seen/head	4,147,176
MLM train — Target7	335,083
MLM train — total	4,482,259
Eval — PPPL	700
Eval — Tatoeba	2,253
Eval — Bible	23,238
Eval — Roundtrip	22,669
Eval — NER	300

4 Method

4.1 Tokenizer Extension

We fix the tokenizer to Glot500-style *append-only injection*:

1. Copy XLM-R's `sentencepiece.bpe.model` (250,002 tokens) as the source.
2. Train a SentencePiece *unigram* auxiliary model on the 99-language mixed corpus ($\alpha = 0.3$ sampling) with `byte_fallback=true` and `character_coverage=1.0`.
3. Select pieces *absent* from the source SPM and append them directly to the SPM protobuf, preserving unigram scores. We do not use HuggingFace `add_tokens()` to stay closer to the Glot500 reference implementation.

The resulting vocabulary expands from 250,002 to 366,666 tokens (116,664 new pieces). Existing token IDs are preserved, but the `<mask>` special token shifts from ID 250001 to ID 366665; we remap it by string identity during initialization (§4.2).

Difference from Yamaguchi et al. Yamaguchi et al. [Yamaguchi et al., 2025] train the auxiliary tokenizer on a small target-only corpus (≈0.01 GB) and append only target-specific pieces, keeping the vocabulary increment small. Our Glot500-style approach trains on 99 mixed language-scripts and appends all 116,664 new pieces. These two designs differ in both training data scope and vocabulary increment size; we fix this axis to Glot500-style and vary only initialization.

4.2 Embedding Initialization

Three rules apply to all five methods: (i) source-tokenizer tokens copy their source row exactly; (ii) only the 116,664 new rows are method-specific; (iii) the `<mask>` row is remapped by string identity, and input embedding and LM head weight tying is maintained. Audit numbers: source length 250,002; target length 366,666; new rows 116,664; `<mask>` copy diff 0.0; LM head tied.

Table 4: Five initialization methods for new embedding rows.

Method	Initialization of new row $\mathbf{e}_j$
`random`	$\mathbf{e}_j \sim \mathcal{N}(0,\, 0.02^2\mathbf{I})$
`mean`	$\mathbf{e}_j = \mathrm{mean}(\mathbf{E}_{\mathrm{src}})$
`fvt`	$S = \mathrm{SrcTok}(\mathrm{surf}(t_j))$; $\mathbf{e}_j = \mathrm{mean}(\mathbf{E}_{\mathrm{src}}[S])$
`weighted_fvt`	$w_s = \lvert\mathrm{piece}(s)\rvert$; $\mathbf{e}_j = \big(\sum_s w_s \mathbf{E}_{\mathrm{src}}[s]\big) / \sum_s w_s$
`family_mean`	$f = \mathrm{family}(\mathrm{prov}(t_j))$; $\mathbf{e}_j = \mathrm{mean}(\mathbf{E}_{\mathrm{src}}[\mathrm{SrcTok}(f)])$

`random` draws from $\mathcal{N}(0, 0.02)$, providing no information. **`mean`** starts at the global centroid of pretrained embeddings—a stable but uninformative starting point. **`fvt`** decomposes each new token's surface form with the source tokenizer and averages the constituent source embeddings, reusing surface/subword information (e.g., a new token `_bambara` decomposed by XLM-R into `[_bam][bara]` starts at the mean of those two rows). **`weighted_fvt`** weights this average by surface-piece length, giving more weight to longer (more specific) source subword pieces. **`family_mean`** replaces surface-form decomposition with a typological prior: it aggregates provenance information from the tokenizer training corpus to determine which language-script each new token most often appears in, maps that language-script to its family, and initializes the new row at the centroid of source embeddings belonging to head languages in the same family.

Family assignment for `family_mean`. Table 5 shows which families Target7 languages belong to and which head source languages `family_mean` can use. For `dtp` (Austronesian), the prior draws from `ind/msa/jav/tgl/mlg`. For `bam` (Mande) and `xav` (Macro-Je), no head language belongs to the same family; `bam` falls back to the nearest Niger-Congo head (`xho`), and `xav` has no usable family prior.

Table 5: Target7 family assignments and head source languages used by `family_mean`.

Target	Family	Head source languages
`dtp`	Austronesian	`ind, msa, jav, tgl, mlg`
`csb`	Slavic	`ces, hrv, pol, slk, slv`
`fur`	Romance	`cat, fra, ita, por, ron`
`lij`	Romance	`cat, fra, ita, por, ron`
`ile`	Constructed	`epo`
`bam`	Mande	Niger-Congo proxy: `xho`
`xav`	Macro-Je	none

4.3 Continued MLM Pretraining

We continue pretraining the initialized model with the MLM objective on the corpus described in §3. All five methods use identical hyperparameters; only the starting checkpoint differs.

Table 6: Continued MLM pretraining settings.

Item	Value
Base model	`xlm-roberta-base`
Objective	MLM (dynamic masking, mask prob. 0.15)
Learning rate	5×10^{-5}
Adam	$\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$
LR schedule / warmup / weight decay	linear / 0 / 0.0
Max sequence length	512
Language sampling	$\alpha = 0.3$
Checkpoint interval	1,000 steps
Convergence budget	50,000 steps (all five methods)

5 Experiments

All five initialization methods share the same tokenizer, corpus, MLM objective, training budget, and evaluation runner. Score differences therefore reflect only the interaction between initialization and training dynamics.

Common sentence embedding. Retrieval and similarity tasks use the Glot500 retrieval-style encoder: hidden states from layer 7 (0-indexed) are mean-pooled over non-special tokens, then L2-normalized. For centered cosine analysis, the global sentence-vector mean is subtracted before normalization.

5.1 Pseudoperplexity (PPPL ↓)

Following **?**, we compute pseudoperplexity over the held-out target sentence pool. PPPL masks each token position in turn and accumulates the negative log-likelihood of the true token:

$$\mathrm{PPPL}(x) = \exp\left(\frac{1}{n}\sum_{t=1}^{n} -\log p_\theta(x_t \mid x_{\setminus t})\right).$$

Lower is better. We use 100 sentences per Target7 language (700 total). Because all five methods use the same tokenizer, PPPL differences across methods cannot arise from tokenizer fertility; they reflect model quality alone.

5.2 Sentence Retrieval — Tatoeba and Bible (↑)

Following **?** and **?**, we embed English and target sentences and retrieve the top-10 nearest neighbors by cosine similarity. A query is a hit if the correct translation appears in the top 10 (Acc@10). Tatoeba provides up to 1,000 English-aligned sentence pairs per language; Bible provides ∼7,000–8,000 verse-aligned pairs per language, making it substantially harder (see Appendix A for analysis of the Bible floor).

5.3 NER (↑)

We follow the XTREME NER protocol [**?**] using WikiANN [**?**]. A token-classification head is attached to the encoder and fine-tuned on English WikiANN for 10 epochs (LR 2×10^{-5}, batch 8×4, max length 256). Zero-shot transfer to the three NER-evaluable Target7 languages (`csb, lij, fur`) is evaluated with seqeval entity-level F1 on BIO-tagged sequences (`O, B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC`).

5.4 Roundtrip Alignment (↑)

Following ?, we assess representation quality without gold labels by computing word alignment through a language cycle and measuring the fraction of source words that return to themselves. We use SimAlign [?] with layer-7 embeddings on the Bible corpus. Each experiment uses five random sets of three intermediate language-scripts and reports the mean.

5.5 Text Classification — Taxi1500 (↑)

We use Taxi1500 [?] for 6-class text classification. A sequence-classification head is fine-tuned on English for 30 epochs (LR 2×10^{-5}, effective batch 16). Because the local Taxi1500 materialization covers only English, we report only the English head score; target-language transfer is marked `not_applicable`.

Alignment with Glot500 protocol. NER learning rate 2e-5 with Adam and English-only fine-tuning matches the Glot500 paper. Text classification effective batch 16 matches Glot500. Retrieval, roundtrip, and PPPL involve no fine-tuning, so initialization and MLM training are the sole sources of score differences.

6 Results

6.1 Tokenizer Fertility

Applying the extended tokenizer to 3,500 Target7 sentences (500 per language) reduces tokens/word from 2.204 to 1.592, a 27.75% decrease. Per-language reductions range from 34.9% (`xav`) and 33.7% (`csb`) down to 12.4% (`ile`). Because all five initialization variants share this tokenizer, subsequent method-to-method differences cannot be attributed to tokenizer fertility.

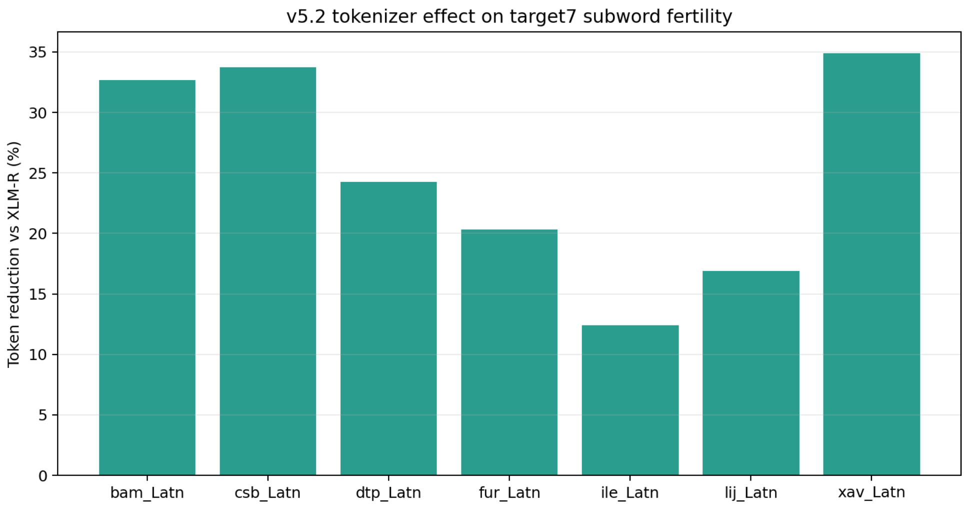

Figure 1: Tokenizer fertility reduction for Target7. The extended tokenizer reduces tokens/word by 27.75% on average. All five initialization methods share this tokenizer.

6.2 50K Downstream Comparison

Table 7 compares all five initialization methods at the 50K checkpoint against the XLM-R-B and XLM-R-L baselines. We follow Glot500's head/tail/all reporting structure.

Table 7: 50K downstream results (head/tail/all). Retrieval and roundtrip ×100; PPPL is raw. "–" denotes missing, pending, or not applicable. **Bold** = best among the five ablation methods.

Metric	Group	XLM-R-B	XLM-R-L	`random`	`mean`	`fvt`	`weighted_fvt`	`family_mean`
PPPL ↓	head	8.1	4.9	23.9	26.3	15.9	**14.6**	18.2
	tail	98.2	63.7	20.2	23.0	16.3	13.9	**12.3**
	all	10.2	6.2	23.6	26.0	15.9	**14.6**	17.6
Tatoeba ↑	head	65.6	54.3	72.1	70.0	72.2	**73.5**	73.3
	tail	20.5	21.6	33.4	33.2	34.4	35.8	**36.1**
	all	63.6	52.8	70.3	68.4	70.5	**71.8**	71.6
Bible ↑	head	38.1	17.3	35.6	39.0	39.6	**39.8**	39.7
	tail	0.5	0.3	0.8	0.8	**0.8**	0.9	0.8
	all	36.7	16.6	34.2	37.5	38.1	**38.3**	38.2
NER ↑	head	62.2	67.5	61.4	62.5	62.7	62.1	**62.8**
	tail	45.7	53.9	48.2	49.2	**51.3**	50.6	49.6
	all	61.6	67.0	60.9	62.0	62.2	61.6	**62.3**
Roundtrip ↑	head	18.5	13.7	19.4	19.6	21.0	**21.1**	19.7
	tail	2.4	2.7	2.7	2.7	3.2	**3.4**	3.2
	all	17.9	13.3	18.7	19.0	20.3	**20.4**	19.1
Text(EN) ↑	head	59.3	72.9	72.6	68.2	**77.4**	74.9	74.2
	tail	–	–	–	–	–	–	–
	all	59.3	72.9	72.6	68.2	**77.4**	74.9	74.2

All five initialization variants substantially outperform the XLM-R-B baseline on tail PPPL and Tatoeba retrieval. Within the ablation, `family_mean` achieves the best tail PPPL (12.3) and Tatoeba tail accuracy (36.1), while `weighted_fvt` leads on head/all PPPL, Tatoeba head/all, Bible, and roundtrip alignment. NER tail is won by `fvt` (51.3); NER head/all by `family_mean` (62.8/62.3). Text(EN) head/all is won by `fvt` (77.4). `random` and `mean` rank lowest overall; `mean` produces the worst PPPL, worse than `random`. Bible tail scores are at floor for all methods (see Appendix A). Changing only initialization shifts target downstream scores, and FVT-family methods that reuse source-tokenizer or typological information consistently outperform uninformed baselines.

6.3 Per-Language PPPL Breakdown for `family_mean`

Table 8 decomposes the `family_mean` tail PPPL advantage by language at 50K.

Table 8: Per-language PPPL at 50K (lower is better). "prior" = number of same-family head languages available to `family_mean`. **Bold** = row minimum.

Language	Family	prior	`random`	`fvt`	`weighted_fvt`	`family_mean`
`dtp`	Austronesian	5	127.4	245.3	157.7	**41.2**
`csb`	Slavic	5	42.7	25.0	**20.5**	20.7
`fur`	Romance	5	54.5	38.5	31.6	**28.8**
`lij`	Romance	5	30.4	22.9	20.2	**18.1**
`ile`	Constructed	1	10.5	7.4	**6.9**	7.1
`bam`	Mande	proxy	144.7	115.9	97.8	**78.0**
`xav`	Macro-Je	none	2.7	2.6	**2.3**	**2.3**

Three patterns emerge. First, `dtp` (Austronesian) accounts for the bulk of `family_mean`'s aggregate advantage. Five Austronesian head languages (`ind/msa/jav/tgl/mlg`) provide a strong family prior, and `family_mean` converges to PPPL 41.2 versus `fvt`'s 245.3—a 6× gap.

Notably, fvt (245.3) is worse than random (127.4) for dtp: the XLM-R source tokenizer decomposes dtp tokens into pieces that carry European semantic signal, making surface-form prior *harmful* relative to random initialization for this language. weighted_fvt (157.7) is likewise worse than random.

Second, Romance (fur, lij) shows clear family_mean gains of 25% and 21% over fvt, while Slavic (csb) and Constructed (ile) show weighted_fvt and family_mean within 0.2 of each other—effectively tied.

Third, even languages without a direct family prior benefit from family_mean. bam (Mande) uses a Niger-Congo proxy (xho) and still achieves PPPL 78.0 versus fvt's 115.9 (33% lower). xav (Macro-Je) has no usable prior; all methods converge to the 2.3–2.7 range with differences below 0.4.

In sum, family_mean's aggregate tail PPPL advantage is concentrated in dtp, confirming the prediction in Table 5: the larger the family coverage in the head set, the greater the benefit of family-aware initialization.

6.4 Loss Convergence

Figure 2 shows MLM training loss for all five methods over 50K steps. Final losses are: weighted_fvt 2.73 $<$ fvt 2.76 $<$ family_mean 2.91 $<$ random 3.12 $<$ mean 3.27. Despite sharing the same tokenizer, corpus, and objective, the five methods converge to distinct loss levels. FVT-family methods reach the lowest loss; mean ends higher than random. Collapsing all new token rows to a single centroid (mean) hinders gradient separation in the early steps, producing persistent disadvantage.

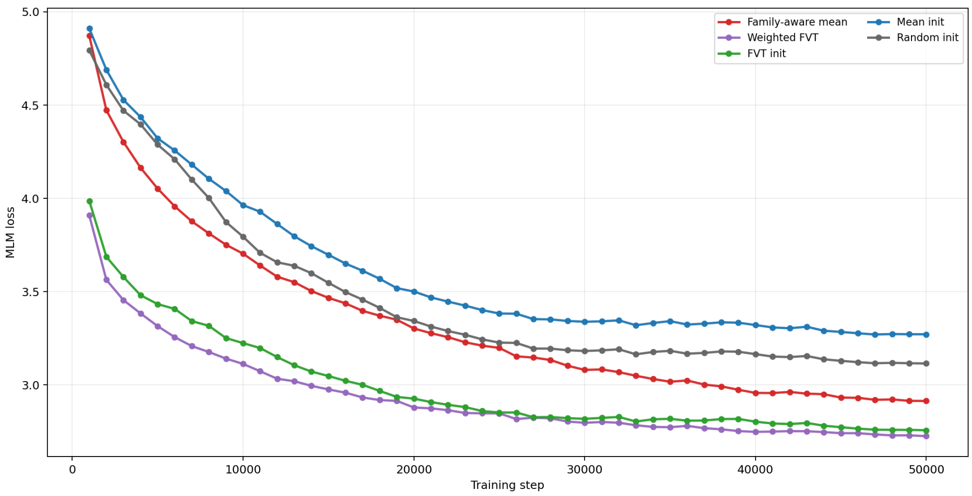

Figure 2: 5-way MLM training loss (50K steps). FVT-family methods converge lowest; mean converges worst. The ranking is stable from early training through convergence.

6.5 Step Trajectory and Crossover

Figure 3 and Table 9 show how tail PPPL and Tatoeba accuracy evolve from 10K to 50K steps.

fvt starts with the best tail PPPL at 10K (21.2) but plateaus thereafter. weighted_fvt and family_mean start lower at 10K (26.9 and 34.2), improve continuously, and overtake fvt by 30K—a clear crossover. random and mean remain at the bottom throughout. **The benefit of a structured initialization prior is visible at convergence, not at early checkpoints.** Evaluating initialization methods from early diagnostics alone would rank fvt first and miss the eventual superiority of weighted_fvt and family_mean. This crossover is consistent with the loss ordering in Figure 2.

Table 9: Step trajectory for tail metrics. PPPL is raw; Tatoeba and Roundtrip ×100.

Method	PPPL (10K/30K/50K)	Tatoeba (10K/30K/50K)	Roundtrip (10K/30K/50K)
`fvt`	21.2 / 16.5 / 16.3	33.9 / 34.3 / 34.4	3.02 / 3.24 / 3.22
`weighted_fvt`	26.9 / 16.1 / 13.9	30.4 / 35.4 / 35.8	2.85 / 3.33 / 3.38
`family_mean`	34.2 / 14.8 / 12.3	30.7 / 35.5 / 36.1	2.61 / 3.09 / 3.16
`random`	28.4 / 20.4 / 20.2	33.0 / 33.2 / 33.4	2.61 / 2.69 / 2.67
`mean`	32.5 / 23.3 / 23.0	31.7 / 32.8 / 33.2	2.68 / 2.73 / 2.72

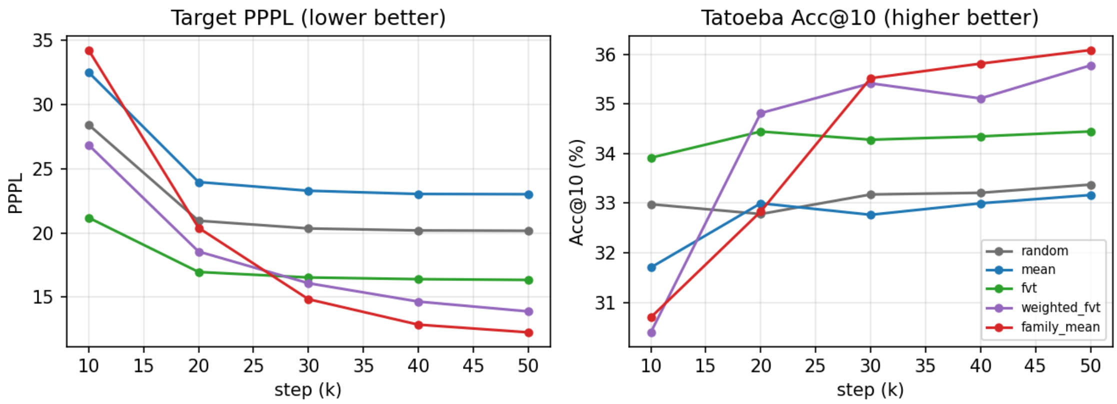

Figure 3: Target metric trajectories from 10K to 50K. Left: tail PPPL (lower is better). Right: Tatoeba tail Acc@10 (higher is better). `fvt` (green) saturates early; `weighted_fvt` (purple) and `family_mean` (red) continue improving and overtake `fvt` by 30K. `random` (gray) and `mean` (blue) remain at the bottom.

7 Representation Analysis

7.1 Sentence Embedding Extraction

We follow the Glot500 retrieval-style encoder protocol. Each sentence is tokenized and passed through `AutoModel`; hidden states from layer 7 (0-indexed, i.e., the 8th layer) are mean-pooled over non-special tokens and L2-normalized. For centered cosine analysis, the global sentence-vector mean is subtracted before normalization. A language centroid is the mean of all sentence vectors for that language.

7.2 Language-Identity and Family/Typology Structure

We ask whether the sentence representation space groups languages by identity and typological family, or merely by script. Since all Target7 languages use Latin script, this analysis is useful precisely because it can separate "same script only" from "same family" relationships. We measure centered cosine similarity between sentence-pair buckets defined by their language relationship.

Table 10 shows results at 50K (`fvt`). Similarity decreases monotonically from same-language pairs (0.455) through same-family tail pairs (0.266), same macro-family (0.146), different-family tail pairs (0.074), and same-family tail–head pairs (0.052), to cross-family Latin pairs (−0.041) and cross-script pairs (−0.063). Crucially, tail–head same-family pairs (0.052) are more similar than tail–head different-family Latin-only pairs (−0.041), confirming that family/typology structure is stronger than script overlap.

Table 10: Centered cosine similarity by relation bucket (fvt, 50K).

Relation	Pairs	Centered cosine
Tail within language	350	0.455
Tail–tail same family	50	0.266
Tail–tail same macro-family	100	0.146
Tail–tail different family	900	0.074
Tail–head same family	1,050	0.052
Tail–head different family, same Latin script	6,100	−0.041
Tail–head different family, different script	1,856	−0.063

7.3 Step-Wise Formation of Clustering

Figure 4 tracks how the relation buckets evolve from 10K to 50K steps under `fvt`. Language-identity and family structure form by 10K and change little thereafter (same language: 0.458 → 0.455; same family: 0.274 → 0.266). Cross-lingual semantic alignment (same-meaning src–eng pairs) shows modest improvement after 10K before saturating. **Easy axes (language identity, local geography) are captured early in training; harder axes (semantic alignment) improve slowly thereafter.**

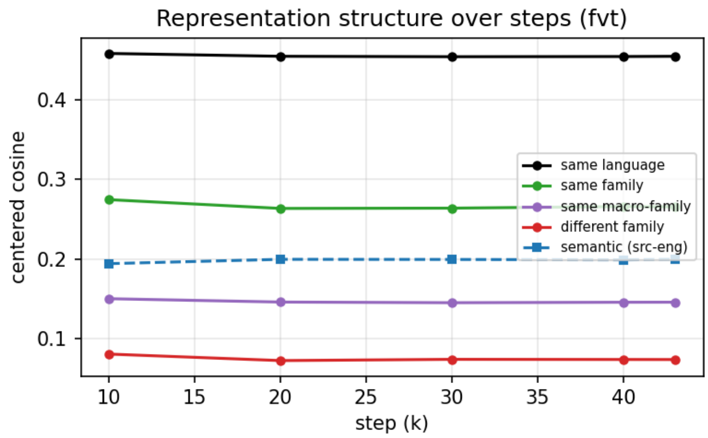

Figure 4: Step-wise change in representation structure (`fvt`). Language and family similarity (solid lines) stabilize by 10K steps; cross-lingual semantic alignment (dashed, src–eng pairs) improves gradually.

7.4 2D PCA Visualization

Figure 5 shows sentence embedding centroids for 38 languages projected to 2D by PCA at steps 10K and 50K under `fvt` (triangles = Target7 tail languages; circles = head/reference languages; color = language family). In both panels, same-family languages cluster together: Slavic `csb` groups with `ces/pol`; Romance `fur/lij` groups with `ita/fra`; Austronesian `dtp` groups with `ind/msa`. The cluster structure is stable between 10K and 50K, consistent with the early formation observed in §7.3. Languages without nearby family members in the head set (`bam, xav`) are relatively isolated. Each tail centroid positions itself toward its family cluster in the head set, consistent with Glot500's related-language help.

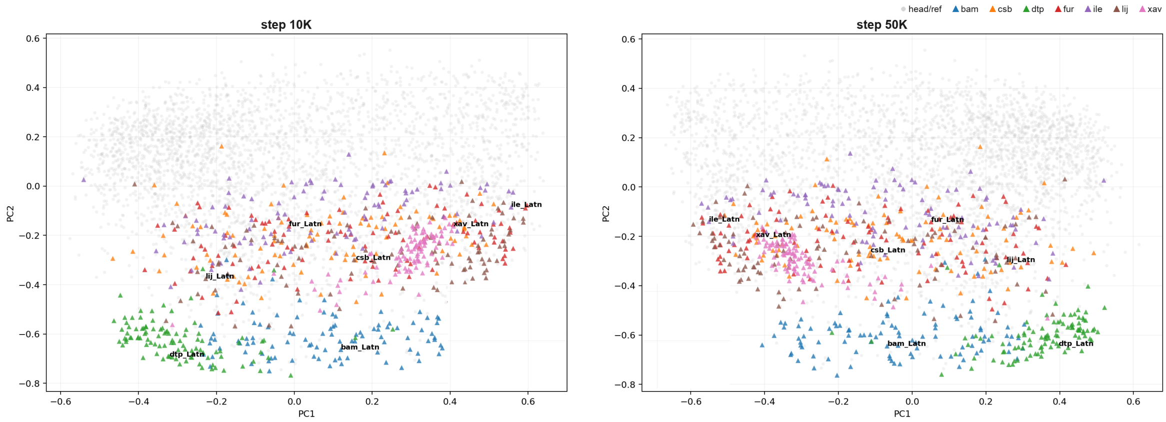

Figure 5: 2D PCA of sentence embedding centroids for 38 languages (`fvt`; left = 10K, right = 50K). Color indicates language family; triangles = Target7 tail languages; circles = head/reference languages. Family clustering forms early and remains stable across steps.

7.5 Synthesis: Which Initialization Is Best?

Combining the representation analysis with downstream results reveals the key mechanism: the model learns to group typologically related languages into nearby regions of the representation space. This is the same direction as Glot500's related-language help—having related languages in the training mix improves tail-language representations [Imani et al., 2023]. `family_mean` encodes this structure as an initialization prior, which is why it converges to the best target PPPL and Tatoeba scores.

Across the five methods: **`family_mean`** is the best choice when tail PPPL and retrieval are the primary targets, especially for languages with strong family coverage in the head set (e.g., `dtp`). **`weighted_fvt`** is the most balanced choice, leading on training loss, Bible head/all, roundtrip alignment, and Tatoeba head/all, while remaining competitive on tail metrics. **`fvt`** shows a "fast start, early plateau" profile: it leads at 10K but fails to improve as `weighted_fvt` and `family_mean` continue to do. **`random`** and **`mean`** are consistently at the bottom; `mean` is worse than `random` on PPPL, confirming that collapsing new rows to a single centroid is harmful.

8 Limitations and Conclusion

8.1 Limitations

This study is a controlled budget ablation; its conclusions are bounded by the following scope.

- **Scale.** The experiment uses 92 head + 7 tail language-scripts, not the 511 language-scripts of full Glot500. Findings may not generalize to a broader language set.
- **Script diversity.** All Target7 languages use Latin script. We make no claim about script diversity or about languages with non-Latin scripts.
- **Family coverage asymmetry.** `family_mean`'s advantage is concentrated in `dtp` (Austronesian), which has five head-language family members. Languages with no usable family prior (`xav`) show no advantage from `family_mean`. Whether family-aware initialization generalizes beyond well-covered families requires a broader head set.
- **Single run.** `weighted_fvt` and `family_mean` are each a single run. Multi-seed replication is needed to confirm that their advantages are stable.

- **Training budget.** The 50K-step window is sufficient to observe crossover and convergence, but longer training may change rankings.
- **Task coverage asymmetry.** NER covers three tail languages; Bible retrieval and roundtrip cover a different three. The aggregate tail scores therefore reflect different language subsets per metric.
- **Text classification.** The local Taxi1500 materialization covers only English, so Text(EN) does not measure target-language transfer.

8.2 Conclusion

We applied Glot500-style vocabulary extension to `xlm-roberta-base` and compared five new-token embedding initialization methods in a controlled 50K-step continued MLM pretraining experiment. Our findings are:

1. **Tokenizer fertility.** The extended tokenizer reduces Target7 tokens/word by 27.75% ($2.204 \to 1.592$), confirming that tail-language fragmentation is substantially reduced.
2. **Initialization affects convergence loss.** Under identical tokenizer, corpus, and objective, the five methods converge to distinct MLM loss values: `weighted_fvt` 2.73 < `fvt` 2.76 < `family_mean` 2.91 < `random` 3.12 < `mean` 3.27. `mean` is worse than `random`, showing that collapsing all new rows to a single centroid is harmful.
3. **Family-aware initialization leads at convergence.** `family_mean` achieves the lowest target PPPL (12.3) and highest Tatoeba tail accuracy (36.1) at 50K, while `weighted_fvt` is the most balanced method across all metrics. The benefit is not visible at 10K—`fvt` leads early—but `weighted_fvt` and `family_mean` overtake `fvt` by 30K and maintain the lead through 50K. Evaluating initialization from early diagnostics alone would yield the wrong ranking.
4. **Family structure in the representation space explains the advantage.** Centered cosine analysis shows that the encoder groups languages by typological family, not merely by script, and that this structure forms by 10K steps and is stable thereafter. `family_mean` places new token rows in the correct family region of the embedding space from the start, shortcutting the alignment process that `random` and `mean` must complete from scratch during training.

In low-resource vocabulary adaptation, embedding initialization is a consequential design choice. Reusing corpus-provenance information (`fvt`, `weighted_fvt`) outperforms random initialization, and explicitly encoding language-family structure (`family_mean`) provides additional benefit for languages with adequate family coverage in the head set. Future work should validate these findings on non-Latin scripts, a wider family distribution, and longer training budgets, and should explore combining surface-form and family priors based on provenance confidence.

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A Artifact and Code Map

Table 11: Reproducibility map for the v5.2 report draft.

Stage	Code / artifact	Role in report
Target selection	`preprocessing/prepare_v5_glot50010_merge_inputs.py`	Defines `V52_OVERLAP_TASKFILL_GLOT5007`; validates XLM-R-unseen status, min 30K sentence count, and raw-dir existence.
Corpus prepare	`preprocessing/run_v52_glot5007_prepare.sh`	Creates target manifest and stats CSV for 92+7 setup.
Corpus merge	`preprocessing/run_v52_glot5007_merge.sh`, `preprocessing/merge_files.py`	Builds sampled MLM corpus with $\alpha = 0.3$, manifest, and JSON report.
Tokenizer	`tokenization/train_v52_glot5007.sh`, `tokenization/run.py`	Trains SentencePiece unigram tokenizer, enables byte fallback, appends new pieces to XLM-R SPM protobuf.

Stage	Code / artifact	Role in report
Initialization	`scripts/run_v52_build_initializers.sh`, `scripts/build_v5_initialized_checkpoint.py`	Builds random/ mean/ FVT/ weighted-FVT/family-aware mean initialized checkpoints; copies rows by token identity and checks mask remap.
MLM training	`modeling/train_v52_glot5007_mlm.sh`, `modeling/run.py`	Runs continued MLM with max length 512, LR $5e-5$, Adam betas $(0.9, 0.999)$, MLM prob 0.15.
Evaluation	`evaluation/retrieval/`, `evaluation/tagging/`, `evaluation/round-trip/`, `evaluation/text_classification/`	Provides PPPL, retrieval, NER/ POS, roundtrip, and text-classification metric families.
Aggregation	`docs/exp/v5.2/3_evaluation/v52_final_downstream_table.tsv`	Source for step-4000 initialization ablation table.
Live status	`docs/exp/v5.2/LIVE_STATUS.md`	Records which initializers/training runs are complete, missing, or running.

B Draft Writing Notes

이 LaTeX draft는 final submission 직전 다음 항목을 확인해야 한다.

- `docs/exp/v5.2/LIVE_STATUS.md`의 latest status로 50K five-way convergence/evaluation 완료 여부를 갱신한다.
- Step-4000 diagnostic table은 `docs/exp/v5.2/3_evaluation/v52_final_downstream_table.tsv` 근거로 유지하되 final claim에 사용하지 않는다.
- head/tail/all final table은 `docs/exp/v5.2/3_evaluation/09_aggregation/main_head_tail_all.tsv` 형식으로 50K checkpoint별 group row를 재생성해 채운다.
- 논문 제출용 PDF에는 [TODO: course name, student id, exact page limit]를 채운다.